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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

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Individual Component Title: Personalized Sign Language Tutoring System for Caregivers

	Identification and Analysis
Data Privacy Concerns	GLOSS is trained on two public collections of isolated American Sign Language (ASL) sign videos. WLASL, introduced by Li et al. at the IEEE Winter Conference on Applications of Computer Vision in 2020, was built from videos found on sign language websites and YouTube. Its release terms, the Computational Use of Data Agreement (C-UDA), limit it to academic and computational use. ASL Citizen, introduced by Desai et al. at NeurIPS in 2023, was collected by Microsoft Research together with Boston University. Its 83,399 videos were recorded on a web platform by 52 Deaf and hard-of-hearing community members. Ethical oversight came from Microsoft as the primary Institutional Review Board of record (#418), with a secondary review at Boston University and a completed data protection impact assessment. Contributors were told how their recordings would be used and released and could delete videos or withdraw before the public release. Signers' faces are visible in both collections, since facial expressions carry meaning in ASL. From the videos of the 59 care-related signs I chose, MediaPipe Holistic extracted landmarks, and only 49 of these, covering the hands and upper body, were retained. Face landmarks were discarded, so the models learn from coordinates alone and never see an image. Both the downloaded videos and the processed files stay in a Google Drive folder that only I can access. They serve this research alone and have not been passed on to anyone. Identifying the signers was never attempted. In the application, each webcam clip is kept only long enough to extract its landmarks and is then discarded. What reaches the database is the outcome of the attempt: the sign recognised, its confidence, the quality tier and the feedback given. Gemini is sent the rule-based feedback text and nothing else, so the video, the caregiver's identity and the model's predictions are not shared with it. No patient records or Parkinson's data set were used for the Parkinson's module, whose content was drawn from published

	clinical sources. Wherever my thesis includes a screenshot, the webcam feed is blurred. Caregivers taking part in the planned evaluation will be given an information sheet and asked for written consent. They will be known only by a code and may stop whenever they wish. Any personal data collected will follow Sri Lanka's Personal Data Protection Act, No. 9 of 2022.
Data Security	All research data and trained models live in Google Drive under my own account and are processed in Google Colab notebooks from that account. To guard against accidental overwriting, original outputs are set to read-only and every rerun saves to its own folder. Everything the application depends on is packaged as a product bundle, and a manifest records a fingerprint for each file. Two parity files, holding ten test sequences and ten complete attempts respectively, allow the deployed application to prove it is running the exact model that was evaluated. Both parity checks passed. Supabase Authentication manages sign-in for the web application, and the GLOSS API rejects any request that lacks a valid access token. Once the token is verified, the server works out which caregiver profile it belongs to and ignores any identifier supplied by the browser, so one user cannot record attempts under another user's name. Tests covered both a missing or forged token and a request carrying a different caregiver's ID. New caregiver accounts can be created only by a single administrator account. Self-registration is not possible, which keeps account management in one place. The Gemini API key and the Supabase service key exist only as server-side environment variables. Neither is committed to the repository or exposed to the browser. Connections among the browser, the API, Supabase and Cloudflare R2 are all made over HTTPS. R2 holds the reference sign videos and the Parkinson's example videos only, and no learner recordings are ever placed there. The database schema changes only through numbered migration files, 0019 to 0024, and a dedicated prefix on the GLOSS tables prevents another component from altering them accidentally.
Bias and Fairness	GLOSS was designed for novice and trained caregivers, and several decisions keep it fair for people who are still learning. The reference signers are fluent adult ASL users, so the execution thresholds were calibrated on correct, fluent signing. This gives every learner the same clear standard to work towards, and instead

	of simply marking an attempt as wrong, the feedback names the side and body part to adjust. A sign counts as mastered only after three strong attempts in a row, so each caregiver moves at their own pace. The macro-F1 of 0.7752 is reported with its 95% confidence interval, and as the data were split by sign rather than by signer, results on new users will be confirmed with Sri Lankan caregivers. Fairness across signs was also considered. Some care signs look alike, such as hurt (F1 0.00) and pain (F1 0.36). To protect learners from an unfair correction, the tutor never names a sign when its confidence is below 0.35 and asks for another attempt instead. When the tutor is confident, it is correct 98.4% of the time. I accepted that 41.8% of repeat requests follow an attempt that was correct, as repetition does less harm to a beginner than wrong feedback. Caregivers who sign slowly or quickly are treated equally, since dynamic time warping compares the shape of a movement rather than its speed. Signing at 0.7 to 1.3 times normal speed moved the distance by only 0.337. GLOSS currently teaches ASL with English instructions. To extend it to caregivers who prefer Sinhala or Tamil and to Deaf residents in Sri Lanka, the next steps are a signer-independent test with Sri Lankan caregivers, recalibration on novice attempts and localisation to Sri Lankan Sign Language.
Transparency and Accountability	The vocabulary, data sources, preprocessing, models, selection rule, execution scoring, feedback design and limitations of this component are all set out in my individual thesis. How the four CareSense components came together as a single application is covered in the group thesis. Three Google Colab notebooks hold the research work, one each for data preparation, model training and selection, and the feedback system. A fixed seed of 42 governs every random step, and the library versions used, among them Python 3.11 and MediaPipe 0.10.14, are written down. Each result reported in my thesis can be traced to a CSV or JSON file in a single evidence folder. Repeating both data preparation and training from scratch produced a retrained ensemble with a test macro-F1 of 0.778, compared with 0.775 originally. Model selection relied solely on validation macro-F1, and each model met the test set only once. On discovering that an earlier prototype had been picked from several runs according to its test score, I changed that rule and documented the change. Bootstrap confidence intervals accompany the results, and the weak signs

	and the approaches that did not work, such as wrist-relative normalisation and a threshold unit mismatch, are reported too. Because the feedback comes from rules that point to the side and body part needing correction, a learner can understand why a sign was marked wrong. The language model is permitted only to reword this text. The code is kept in the shared GitHub repository, every screen of the Parkinson's module states that it does not diagnose, and the thesis appendix discloses my use of AI tools. Responsibility for the work remains mine.
Impact on Stakeholders	**Caregivers and trainees.** They are affected most directly. After training, GLOSS gives them an inexpensive way to keep practising care signs, with feedback that tells them what to change. A wrong correction or frequent repeat requests could discourage a learner, so the tutor asks for a repeat instead of naming a sign it is unsure of. GLOSS is meant to support practice between lessons, not to replace a qualified teacher. Its requirements came from trainees at the Colombo Caregiver and Health Science School, who said they remember only a few signs once training is over, and from a sign language tutor, who advised that varied signing speeds should be accepted. **Deaf and hard-of-hearing residents.** They may find it easier to express basic needs to caregivers. My thesis makes clear that GLOSS teaches ASL rather than Sri Lankan Sign Language and does not replace interpreters. **Residents with Parkinson's disease.** Changes in their movement may be noticed and reported sooner. A caregiver could mistake normal ageing for a symptom, so every screen of the module asks them to observe, record and report to clinical staff, never to diagnose. The module was added on the advice of the chairperson of Remark Elder Care. **Care homes and training schools.** They gain an affordable training tool. Because the data set licences restrict commercial use, a licensed reference set recorded with Deaf signers will be needed before any paid use. **Data set creators and contributors.** The creators and signers of WLASL and ASL Citizen are respected by following their terms of use, not redistributing their videos and acknowledging them in my thesis. This work also contributes to Sustainable Development Goals 3, 4 and 10: Good Health and Well-being, Quality Education and Reduced Inequalities.